

Project Proposal (total 100%)

Project Name (2%): X_swarm

Project Team Members (4%):

Emmett Gilbert

Gatlin Nelson

DeAndre Johnney

Nicholas Schultz

Problem Statement (24%):

What is the problem you are planning to solve?

We are trying to solve the issue of heterogenous aerial swarm formations with ground vehicles.

Why is it a problem?

Coordinating flying vehicles with ground vehicles is difficult because of network throughput, delay, computational effort, and other software/hardware inaccuracies.

What type of constraints the problem have? [eg: physical, cost, schedule, etc.]

This problem has software and testing constraints, which ultimately lead to time constraints.

Stakeholders (10%):

Who are the stakeholders?

Primary stakeholders: Equal share amongst all stakeholders, 25%.

Secondary stakeholders:

Proposed Solution (45%):

This defines the scope of the proposal.

What do you plan to cover, and what is out of scope?

What standards drive the solution? [eg, IEEE standard, FAA, FCC, ISO, etc.]

This is where you start identifying system functionalities that will be complete by the end of the semester and complete by the end of the year.

This describes your approach to the solution. For example, if you are planning to have an interactive map that displays specific information, if you will use a computer vision with camera to identify objects, if you plan to have the actual real data or simulated data, etc.

Semester 1: Testing and research, avoiding simulations as much as possible to gain real time experience.

Semester 2: Implementation

Proposed Project Budget (10%):

How much money is required to complete your project? Make a list of costs and what they are associated with, if applicable. *Note: If there is no budget required, state it here, and the 10% will be split to 3% problem statement, 3% stakeholders, and 4% proposed solution)*

References (5%):